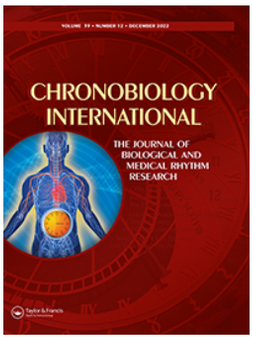

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ORIGINAL ARTICLE

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Effects of long-haul transmeridian travel on physiological, sleep, perceptual and mood markers in Olympic team support staff

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ABSTRACT

The objective of this study was to holistically examine the impact of long-haul transmeridian travel (LHTT) on perceptual, mood, sleep and physiological markers in Olympic team support staff travelling to Japan for the 2020 Summer Olympic Games. An observational descriptive study design was used. Nine support staff members of the Irish Olympic team (2 M/7 F; age 34.3 ± 8.3 y (mean ±SD)) embarked on a long-haul (LH) eastward flight across eight time-zones from Ireland to Japan (approx. 24 h total travel time), to work at the Irish Team's 2020 Summer pre-Olympic Games camp, postponed to July 2021 due to Covid-19 pandemic. Perceived jet lag and travel fatigue symptoms, mood states and salivary markers for circadian rhythm and stress were assessed in the morning and evening during the week prior to travel as baseline (BL) measures and on days 1 to 8 (C1-C8) and day 15 (C15) post-travel. Night-time sleep (duration and quality) was monitored via actigraphy monitors and self-report sleep diaries. Participants perceived themselves to be significantly jet lagged for six days post-travel ($p < .05$). Morning sCort decreased by 66% on C1 and remained significantly lower than BL until C6 ($p \leq .03$). On arrival participants perceived sleep to be worse than BL on arrival (C1, C2, C4, C5; $p \leq .04$), with significantly shorter sleep duration (C2, C3, C6; $p \leq .01$) and lower sleep efficiency (C2, C6; $p \leq .04$) recorded by actigraphy, all normalizing by C7. Negative changes in mood states were evident in the evening time following LHTT, with significant elevations in confusion (C2, C3, $p \leq .02$), fatigue (C2, C3; $p \leq .03$) and depression (C3, C7; $p < .05$) and reduction in vigour (C2, C6, C7; $p < .05$). Following LHTT in an eastward direction across eight time-zones, it took seven days for perceived jet lag, physiological markers for circadian rhythm and sleep to normalize in Olympic team support staff. Despite alleviation of jet lag and fatigue and return of sleep to normal by C15, vigor remained low, indicating a "submerged" mood profile in these Olympic team support staff. These findings highlight the need to put strategies in place before and after LHTT for the Olympic Games to assist Olympic team support staff to maximize sleep, minimize stress and assist with expediating recovery from jet lag and travel fatigue, allowing them to perform optimally in supporting Olympic athletes in their final preparations for the Games.

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Introduction

Support staff working in elite sport frequently embark on long-haul transmeridian travel (LHTT) for training and major international competitions such as the Tokyo Summer Olympic and Paralympic Games in 2021 and the Beijing Winter Olympic and Paralympic Games in 2022. In general, "travel" has been identified as an organizational stressor experienced by high-performance coaches (Thelwell et al. 2008). Additionally, LHTT across three or more time zones typically causes travel fatigue and jet lag, and is associated with physical and physiological decrements, cognitive and psychomotor disturbances, disrupted sleep, and increased risk of illness (Reilly 2001; Reilly et al. 1997; Schwellnus et al. 2012); all of which could prevent optimal performance of elite athlete support staff.

In the elite sporting environment, the health, well-being, and performance of support staff are essential to successful athlete support and ultimately athlete performance. Support staff for elite athletes frequently experience a high degree of strain, caused by the physical and emotional stress of the role and the complex and unpredictable nature of the work environment. This may result in support staff presenting with symptoms associated with recovery-stress imbalance, which can be exasperated by travel and time-zone changes (Calleja-González et al. 2021). A case study examining sleep in a football team and their head coach observed that the

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sleep/wake behavior of the coach was poorer than the athletes following LHTT, although bedtime was earlier, there was less time spent in bed, shorter sleep duration, lower sleep efficiency and earlier wake up times (Lastella et al. 2017). Following LHTT, facilitating a smooth transition for athletes to the new training/competition environment will be a priority objective for support staff, thus LHTT for training/competition presents a situation of heightened workload and strain for support staff. By its very nature, the Olympic environment is particularly stressful and unlike any other in terms of media attention (Olusoga et al. 2012), so will further intensify workload and strain in Olympic team support staff (Terry et al. 1997).

The symptoms of travel fatigue and jet lag are numerous and complex and often difficult to separate (Atkinson et al. 2014; Bin et al. 2019). Travel fatigue is associated with symptoms of general fatigue, disorientation and weariness that occur during or immediately following long-haul (LH) travel in any direction (Samuels 2012; Waterhouse et al. 2007, 2004) . Travel fatigue is caused by challenges of LH travel, including situations that the support staff may perceive as stressful (Waterhouse et al. 2007, 2004), such as directing a team through the airport, flight delays, missing baggage and customs clearance. Travelling support staff generally experience disrupted sleep duration and quality due to travel schedules (Lastella et al. 2019) and cramped cabin conditions (Roach et al. 2018; Sack 2009) and have limited access to preferred food and fluid choices due to airline restrictions and airline meal schedules (Reilly et al. 2007). Poor air quality, low relative humidity and "cabin altitude" on the aircraft itself can also contribute to travel fatigue (Waterhouse et al. 2007, 2004) and risk to health (Schwellnus et al. 2012). Travel fatigue can be acute following any individual long journey, or chronic (cumulative) because of repetitive travel within a season (Janse van Rensburg et al. 2021; Samuels 2012). Recovery from travel fatigue is expected to take a day or two with adequate rest, sleep, and nutrition (Sack 2009; Samuels 2012; Waterhouse et al. 2007, 2004).

A recent consensus statement has defined jet lag as a "temporary impairment of sleep and wakefulness, as well as other biological functions, associated with rapid eastward or westward travel across 3 or more time-zones" (Janse van Rensburg et al. 2021, 2032). Jet lag can cause an array of symptoms (Simmons et al. 2015), including poor sleep, gastrointestinal disturbance, negative mood disturbances, and impaired physical and cognitive performance (Forbes-Robertson et al. 2012; Sack 2009; Simmons et al. 2015; Waterhouse et al. 2007, 2004). Symptoms of jet lag are generally more severe and last longer following an eastward flight in comparison to a westward flight (Reilly et al. 1997; Roach and Sargent 2019; Simmons et al. 2015) and when greater number of time-zones are crossed (Reilly et al. 1997; Sack 2009). The human body clock needs time to gradually adjust and consequently the symptoms of jet lag will persist until circadian rhythms shift to a new environment (Sack 2009; Simmons et al. 2015; Waterhouse et al. 2004).

In recent years, there is increased awareness of the potential negative consequences of LHTT on elite athlete health and performance and guidelines to optimize recovery from LHTT and minimize its negative effect on athlete health, wellbeing, and performance have been published (Janse van Rensburg et al. 2021; Roach and Sargent 2019) There are indications that these guidelines are effective, as recent studies show that although elite athletes perceive themselves to be jet lagged and experience some disruption to various physiological mechanisms following LHTT, the effect of LHTT on subjective sleep and physical performance are minimal (Rossiter et al. 2022, 2022). However, no such guidelines exist for support staff travelling with athletes, who experience similar challenges of LHTT as athletes but are generally required to engage in complex decision-making on arrival, with little or no opportunity to recover from travel fatigue and jet lag.

This novel study aimed to assess the impact of LH eastward travel across eight time-zones on perceptual measures, mood, sleep, and biomarkers associated with circadian rhythm and stress in Olympic team support staff travelling to Japan for the 2020 Tokyo Summer Olympic Games.

Materials and methods

Participants

Nine Olympic team support staff (2 M/7 F; 34.3 ± 8.3 y (mean ± SD)) who were members of the Irish Olympic team for the Tokyo Olympics, volunteered to participate in this study. The group consisted of three physiotherapists, two administrators/translators, two performance managers, one strength and conditioning coach and one physiologist. The 2020 Tokyo Summer Olympic Games were postponed by one year due to the Covid-19 pandemic, thus in July 2021, all participants embarked on a LH eastward flight across eight time-zones from Ireland to Japan (approx. 24 h total travel time), to work at the Irish Team's Olympic Games Pre-Games camp. Prior to data collection, all participants provided written informed consent. This study was approved by the University's human research ethics committee and

conformed to the declaration of Helsinki pertaining to research using human subjects.

Study design

For this study, an observational descriptive design was used. Baseline data (BL) were collected three and two days before leaving Ireland, BL1 and BL2, respectively. On arrival to Japan, data was collected every day for eight days after arrival (C1-C8) and again fifteen days after arrival (C15) (Table 1).

Participants travelled from Dublin, Ireland to Tokyo, Japan on one of two routes. Six participants embarked on Route 1 and three participants embarked on Route 2. Details of the routes, departure times, arrival times and travel durations are provided in Table 2. On arrival to the airport in Tokyo, all participants were required to take a Covid-19 test and wait at the airport for the results to processed. On receiving a negative Covid-19 result, participants then proceeded through airport immigration and customs. All participants then travelled by private bus from the airport in Tokyo to the Pre-Games Camp accommodation in Fukuroi City. In July, the journey from Ireland to Japan required an eight-hour eastward time-zone change.

Prior to departure, travel, sleep, and light exposure recommendations specific to their journey were provided to participants via workshops and infographics, however there was no monitoring of these recommendations during travel. The participants did not attempt to shift their circadian rhythm prior to departure through adjustment of sleep/wake schedule or the use of melatonin. The participants refrained from the use of melatonin and hypnotic medications for the duration of the study period. Main meals were provided by the hotel at fixed times. Due to the diverse nature of their roles and unplanned, reactive nature of their work, daily schedules were set at the participants' own discretion and no attempt was made to record schedule of daily activities.

Procedures

Physiological markers

Participants collected a saliva sample immediately on rising in the morning (AM) and just before bedtime (PM) at BL1, BL2, C1-C8 and C15. They were required to avoid eating, brushing their teeth, chewing gum and physical activity for at least one hour before providing the saliva sample. Salivary samples were collected using an oral fluid collector kit (Soma Bioscience, Wallingford, UK), in accordance with manufacturer's guidelines. Samples were analyzed using a lateral flow device (LFD) with the same cartridge used to analyze salivary cortisol (sCort) and salivary alpha-amylase (sAA).

Sleep markers

Sleep actigraphy data were collected for 11 nights that comprised of a two nights baseline (BL1, BL2) and eight nights starting on the night of arrival to Japan (C1-C8) and again on night 15 post-arrival (C15). Sleep/wake behavior was monitored using self-report sleep diaries, wrist activity monitors (Actiwatch Spectrum, Philips Respironics, Murrysville, PA) and perceived sleep data via Liverpool John Moore's University Jet Lag Questionnaire (LJMUQ, Waterhouse et al. 2000). Activity monitors have been found to be a valid method of monitoring sleep in elite athletes, with very good agreement rates for sleep-wake thresholds (81–90%) with polysomnography, the gold standard for measuring sleep (Sargent et al. 2016). Participants were instructed to put on the activity monitor before bedtime and were permitted to remove the monitor on rising the following morning. The bedtime and get-up times recorded in the sleep diaries and data from the activity monitors were used to determine the amount and quality of sleep participants obtained. This was achieved through the Philips Respironics Actiwatch algorithm where time was scored as awake unless (1) the sleep diary indicated the participant was lying down attempting to sleep and (2) the activity counts derived from the activity monitor were sufficiently low to indicate that the participant was immobile (i.e., where the weighted activity count from an epoch fell below the defined threshold). In this study, the scoring process was conducted with a sensitivity set at "medium" which corresponds to a threshold activity count of 40. Once these conditions were met simultaneously, time was scored as sleep. This algorithm has been used in the literature to quantify the sleep/wake behavior (Sargent et al. 2014). The following sleep variables were extracted from the activity monitors and the sleep diary data:

- Bedtime (hh:mm): the self-reported clock time at which the athlete went to bed to attempt to sleep.
- Get-up time (hh:mm): the self-reported clock time at which the athlete got out of bed and stopped attempting to sleep.
- Time in Bed (h): the amount of time spent in bed attempting to sleep between bedtime and get-up time.
- Total sleep time (h): the total amount of sleep obtained during a sleep period.
- Sleep latency (min): the period between bedtime and sleep onset time.

Table 1. Schematic representation of study design.

Study day	1	2		3		4	5	6		7		8		9		10		11		12		13		14		15	16	17	18	19	20		21	
Study Phase		BL1		BL2						C1		C2		C3		C4		C5		C6		C7		C8									C15	
		Home, Ireland					Travel days			Pre-Games Holding Camp, Fukuroi City, Japan																								
		AM	PM	AM	PM		AM	PM	AM	PM	AM	PM	AM	PM	AM	PM	AM	PM	AM	PM	AM	PM	AM	PM							AM	PM	AM	PM
Sleep Log		✓		✓					✓		✓		✓		✓		✓		✓		✓		✓										✓	
Sleep Actigraphy	✓	✓	✓	✓				✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								✓	✓		
LJMUQ (jet lag, fatigue, sleep)		✓		✓					✓		✓		✓		✓		✓		✓		✓		✓										✓	
LJMUQ (jet lag, fatigue, function, meal, bowel)			✓		✓					✓		✓		✓		✓		✓		✓		✓		✓										✓
BRUMS		✓	✓	✓	✓				✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓									✓	✓
Salivary Markers		✓	✓	✓	✓				✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓									✓	✓

BL, Baseline; C, Pre-Olympic Games camp; C1 is camp day 1 and so on; LJMUQ, Liverpool John Moore's University Jet Lag Questionnaire; BRUMS, Brunel Mood Scale

Table 2. Route 1 and Route 2 travel details for the journey from Dublin, Ireland to Fukuroi City, Japan.

	Flight 1	Flight 1 departure time	Flight 1 arrival time	Flight 1 duration	Layover Duration	Flight 2	Flight 2 departure time	Flight 2 arrival time	Flight 2 duration	Airport Layover Duration	Bus departure time	Bus journey duration	Bus arrival time	Total Travel Time
Route 1	DUB-LHR	10:45	12:15	1.30	3.15	LHR-HND	15:30	10:55	11.25	3:55	14:40	3:30	18:10	23.35
Route 2	DUB-DOH	15:00	00:15	7.15	1.55	DOH-NRT	02:10	18:40	10.30	3:40	22:00	4.20	02:20	27.20

Dub = Dublin; LHR = London; DOH = Doha; HND = Tokyo; NRT = Tokyo

- Sleep efficiency (%): sleep duration expressed as percentage of time in bed.
- Perceived sleep: assessed via LJMUQ (described below)

Perceptual markers

Following AM or PM saliva collection, the LJMUQ was completed on BL1 and BL2, and during the Pre-Games Camp (C1 – C8, C15). Subjective ratings of jet lag were obtained on a visual analog scale with a score of 0 = no jet lag and 10 = very bad jet lag. The questionnaire also assessed sleep (5 items; sleep latency, quality, inertia, sleep-time and waking time), fatigue (1 item), mental performance (3 items; concentration, motivation, and irritability), appetite (3 items; hunger prior to meal, meal palatability, and post-meal satisfaction) and bowel function (2 items; frequency and stool consistency). Ratings of these components were measured on a – 5 to + 5 scale with 0 representing 'normal' habitual ratings prior to travel. Only jet lag and fatigue were assessed in both AM and PM questionnaires, all other components were assessed either in the AM (sleep) or PM (mental performance, appetite, bowel function). Using a method described elsewhere (Thompson et al. 2013), the questionnaire data were subsequently pooled into 5 components (jet lag, sleep, fatigue, mental function, appetite, and bowel function). Some of the questions were on a linear scale with lower scores meaning worse than normal symptoms and higher score meaning better than normal symptoms or vice versa. Conversely, some variables had no positive response. Thus, data with a negative outcome were given a positive score and data with a positive outcome were given a negative score. These were then summed and the greater the value of the overall constructs the worse the related symptoms. A recent consensus statement considers that LJMUQ to be one of the more accurate and reliable subjective tools to assess different aspects of jet lag (Janse van Rensburg et al. 2021). LJMUQ has been demonstrated to be highly internally reliable (Cronbach's alpha = 0.85) and internally consistent (Ledger et al. 2020), however it has not been validated against objective measures of circadian disruption.

Mood markers

In the AM and PM (BL1-2, C1-C8 and C15), participants completed the 24-item Brunel Mood Scale (BRUMS, Terry et al. 1999), which was used to assess mood responses. BRUMS is a derivative of the Profile of Mood States (POMS) and is an extensively used psychometric tool to assess mood states in the sport and exercise psychology domain (Terry et al. 2021). BRUMS has six subscales (tension, depression, anger, vigor, fatigue, and confusion) of four mood descriptors each. The participants were prompted to rate their moods based on how they feel "right now" on a 5-point Likert-type scale anchored by 0 = not at all and 4 = extremely. Total subscale scores range from 0 to 16. Raw scores are transformed into standard scores (T-scores) with reference to established tables of normative data for male and female non-athletes (Terry et al. 2021).

Statistical analysis

Data are presented as mean ± SEM. Descriptive statistics were calculated for all variables using MLwiN software (version 3.04, Centre for Multilevel Modelling, University of Bristol). Data were analyzed using a multi-level modelling approach (Rasbash 2009). Multi-level modelling was used as there were multiple testing time points and the study sought to investigate both between- and within-subject variability. A two-level model was conducted, study phase (level 1) and participants (level 2); and two levels were created for each variable separately. This was a random intercept model where repeated observations and participants were assumed to be normally distributed. These assumptions were assessed using a Q-Q plot and it was confirmed that these assumptions were acceptably normal (Goldstein 1991). Analysis was used to identify changes in mean values of all variables across the identified time points. No significant difference was found between BL1 (day 1) and BL2 (day 1), therefore BL1 and BL2 were averaged to give a mean baseline (BL). Significance was set at $p < .05$.

Results

Perceptual markers

Self-reported jet lag and fatigue AM and PM values are presented in Figure 1 (a) and (b), respectively. Perceived jet lag AM was significantly elevated in comparison to BL for six days (C1-C6; $p < .05$) post-travel, reducing to insignificant levels on the morning of C7 (beta = 0.559 [95% CI −0.90 to 2.01]; $p = .45$). In the evening times jet lag PM was significantly elevated for C1-C4 and C6 ($p < .05$). Perceived fatigue was similar to BL in the morning times (C1-C8 and C15) but was significantly higher than BL on three evenings during the camp (C1, C2 and C6; $p > .05$). The highest levels of fatigue (beta = 2.51 [95% CI 1.32 to 3.7]; $p < .001$) and jet lag (beta = 4.836 [95% CI 3.8 to 5.9]; $p < .001$) were reported on the evening of C2.

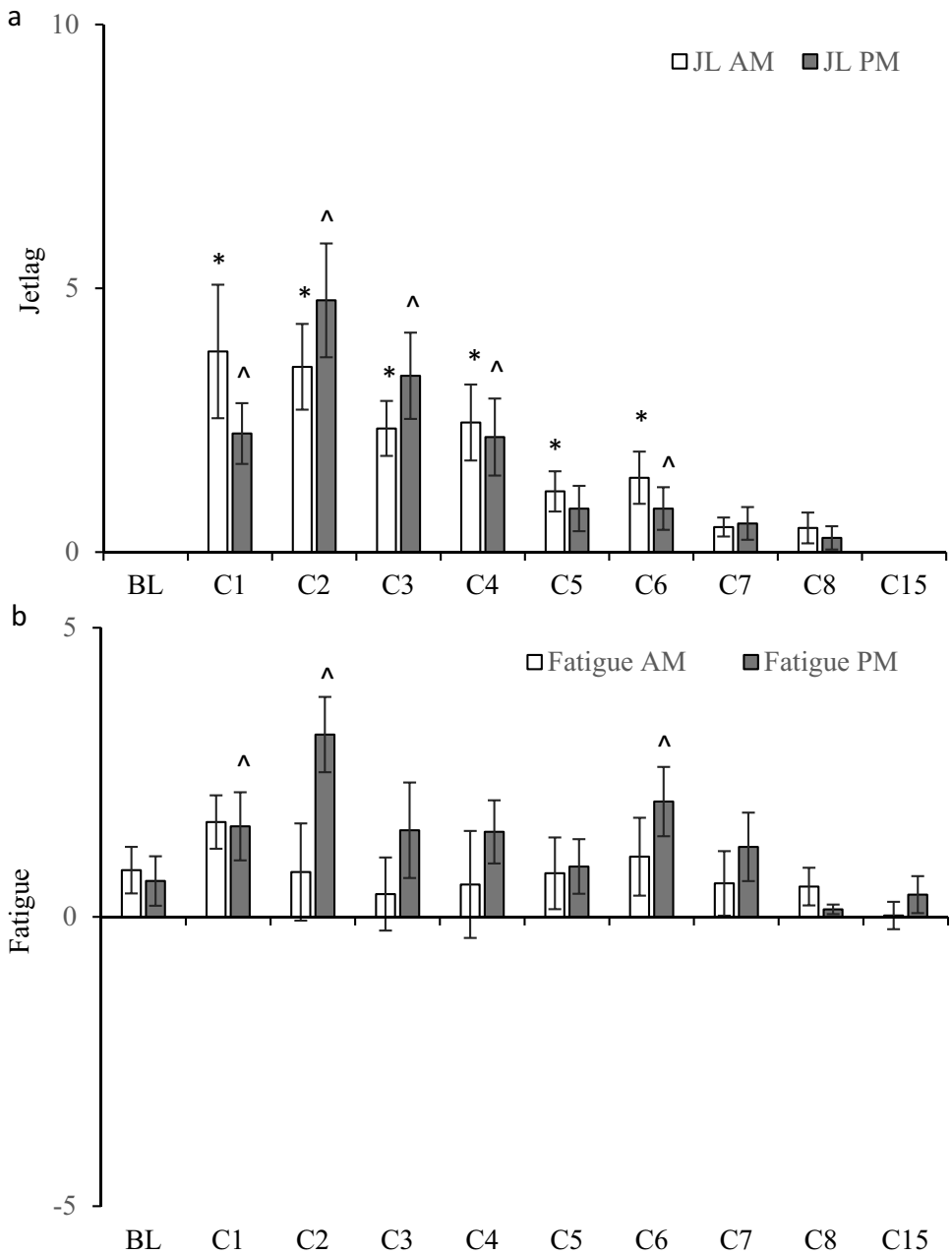

Figure 1. Liverpool John Moore Jet Lag Questionnaire. Perceived jet lag (a) measured at baseline (BL) and post-travel in the morning (JL AM) and evening (JL PM), and fatigue (b) values measured at baseline (BL) and post-travel in the morning (Fatigue AM) and evening (Fatigue PM), during days 1 to 8 (C1-C8) and day 15 (C15) of pre-Olympic Games camp. Higher values indicate worse symptoms. Data presented as mean ± SE. * indicates statistical significance ($p < .05$) in comparison to morning scores at BL determined by multi-level regression analysis. ^ indicates statistical significance ($p < .05$) in comparison to evening scores at BL determined by multi-level regression.

Perceived mental performance was significantly disturbed in comparison to BL on C2, C3 and C7 ($p < .02$). In comparison to BL, there were also significant disturbances to appetite (C1, C3, C5; $p < .05$) and bowel function (C3, C6; $p < .05$) reported following LHTT.

Sleep markers

Sleep data recorded by sleep diary, actigraphy and subjectively via LJMUQ are presented on Table 3. Participants went to bed significantly earlier on C1, C3, C4, C7 ($p \leq .004$) and got up significantly earlier (C1-C8 and C15; $p \leq .03$) during the camp in comparison to BL values recorded at home. Although participants perceived the poorest night's sleep to occur on the night of arrival (C1; beta = 4.85 [95% CI 2.51 to 7.19]; $p < .001$), objectively the worst night's sleep was on C2 with the earliest get up time (beta = −2:17 [95% CI −2:53 to −1:41] h:mm; $p < .001$, shortest time in bed (beta = −1.8 [95% CI −2.9 to −0.7] h; $p < .001$), shortest sleep time (beta = −1.8 [95% CI −2.8 to −0.8] h; $p < .001$) and lowest sleep efficiency (beta = −8.6 [95% CI −16.0 to −1.2] %; $p = .01$). In comparison to BL, significant disturbances in sleep continued, with significant reductions in time in bed (C3, C5, C6; $p \leq .03$), total sleep time (C3, C6; $p \leq .01$), sleep efficiency (C6; $p = .04$) and perceived sleep (C4, C5; $p \leq .04$). On C15, the participants woke significantly earlier than BL (beta = 00:39 [95% CI −1:18 to 0:00] h:mm; $p = .03$) but sleep onset latency was significantly shorter (beta = −16.8 [95% CI −33.0 to 0.7] min; $p = .02$) and total sleep duration was significantly longer (beta = 0.8 [95% CI −0.1 to 1.7] h; $p = .04$).

Physiological markers

sCort and sAA (AM and PM) concentrations at each time point are presented in Figure 2 (a) and (b), respectively. In comparison to BL, there was a 66% decrease in sCort AM on C1 (beta = −5.0 [95% CI −8.0 to −2.0] $ng.ml^{-1}$; $p < .001$) and sCort AM remained significantly lower than BL from C1 to C5 ($p \leq .03$). On the morning of C6, sCort AM increased to levels similar to BL (beta = 0.75 [95% CI −1.87 to 3.38] $ng.ml^{-1}$; $p = .29$). On C7, sCort AM increased above BL levels (61%; beta = 4.5 [95% CI 1.7 to 7.4] $ng.ml^{-1}$; $p < .001$) but sCort AM was similar BL levels on C8 (beta = 0.9 [95% CI −2.0 to 3.7] $ng.ml^{-1}$; $p = .29$) and C15 (beta = 2.2 [95% CI −0.8 to 5.2] $ng.ml^{-1}$; $p = .07$). sCort PM values remained similar to BL throughout the camp.

Apart from elevated sAA on the evening of C3 (beta = 225.9 [95% CI 49.6 to 402.1] $ng.ml^{-1}$; $p = .01$), there were no other significant differences in sAA AM or PM values measured on camp in comparison to BL.

Mood markers

Table 4 represents standard scores (T-scores) for each of the six mood subscales for the duration of the study. There were positive changes in mood state found in the morning times following LHTT in comparison to BL, with significant reductions in confusion (C4, C7, C8, C15; $p \leq .03$), depression (C1, C2, C15; $p \leq .03$) and increase in vigour (C7; $p = .02$). However, negative changes in mood state were more prevelant in the evening times following LHTT, with significant elevations in confusion (C2, C3, $p \leq .02$), depression (C3, C7; $p < .05$) and fatigue (C2, C3; $p \leq .03$) and reduction in vigour (C2, C6, C7; $p < .05$) in comparison to BL. Following LHTT, there was evidence of alleviation of tension in the morning (C3, C5-C15; $p \leq .03$) and evening times (C1, C4, C7-C15; $p \leq .03$) following LHTT. There was a significant reduction of fatigue in comparison to BL in the evening times towards the end of the camp on C8 and C15 ($p < .05$).

Table 3. Sleep/wake behaviours at Baseline (BL) and during holding camp (C1-C8 and C15) recorded objectively by actigraphy and subjectively by Liverpool John Moore's University Jet Lag Questionnaire.

	Sleep Actigraphy						LJMUQ
	Bedtime (hh:mm), LT	Get up Time (hh:mm), LT	Time in Bed (h)	Total Sleep Time (h)	Sleep Latency (min)	Sleep Efficiency (%)	Perceived Sleep
BL	23:29 (00:15)	07:44 (00:21)	8.3 (0.4)	6.8 (0.2)	33.3 (9.5)	83.4 (2.4)	2.6 (0.6)
C1	21:18 (00:43)*	05:27 (00:19)*	8.0 (0.6)	7.0 (0.6)	14.9 (2.6)*	86.9 (3.5)	7.5 (1.8)*
C2	22:47 (00:19)	05:11 (00:18)*	6.4 (0.4)*	5.0 (0.7)*	18.9 (3.8)*	74.8 (7.5)*	5.1 (1.0)*
C3	22:35 (00:21) *	05:38 (00:20)*	7.1 (0.5)*	5.7 (0.3)*	24.3 (7.6)	82.0 (3.6)	3.4 (1.5)
C4	22:38 (00:20) *	06:21 (00:44)*	7.7 (0.7)	6.3 (0.6)	26.3 (7.1)	82.2 (3.1)	4.9 (1.4)*
C5	23:17 (00:26)	06:27 (00:25)*	7.3 (0.4)*	6.4 (0.4)	17.8 (3.1)*	87.2 (1.6)	4.6 (1.6)*
C6	23:26 (00:27)	06:30 (00:18)*	6.8 (0.5)*	5.5 (0.4)*	34.1 (8.7)	80.5 (3.5)*	4.4 (1.1)
C7	22:22 (00:20)*	06:57 (00:15)*	8.6 (0.2)	7.5 (0.2)	22.3 (5.8)	87.0(1.4)	3.6 (1.3)
C8	22:45 (00:23)	07:02 (00:25) *	8.3 (0.2)	6.9 (0.3)	21.4 (11.8)	83.6 (2.7)	2.3 (0.8)
C15	22:49 (00:22)	07:04 (00:14) *	8.7 (0.8)	7.6 (0.7)*	16.5 (5.2)*	87.0 (1.9)	2.6 (0.9)

BL, baseline; C, pre-Olympic Games camp; C1 is camp day 1 and so on; LJMUQ, Liverpool John Moore's University Jet Lag Questionnaire; * indicates statistical significance ($p < 0.05$) in comparison to BL; all values are expressed as mean (SE); all times are expressed as local-time (LT); higher values for Perceived Sleep indicate worse symptoms.

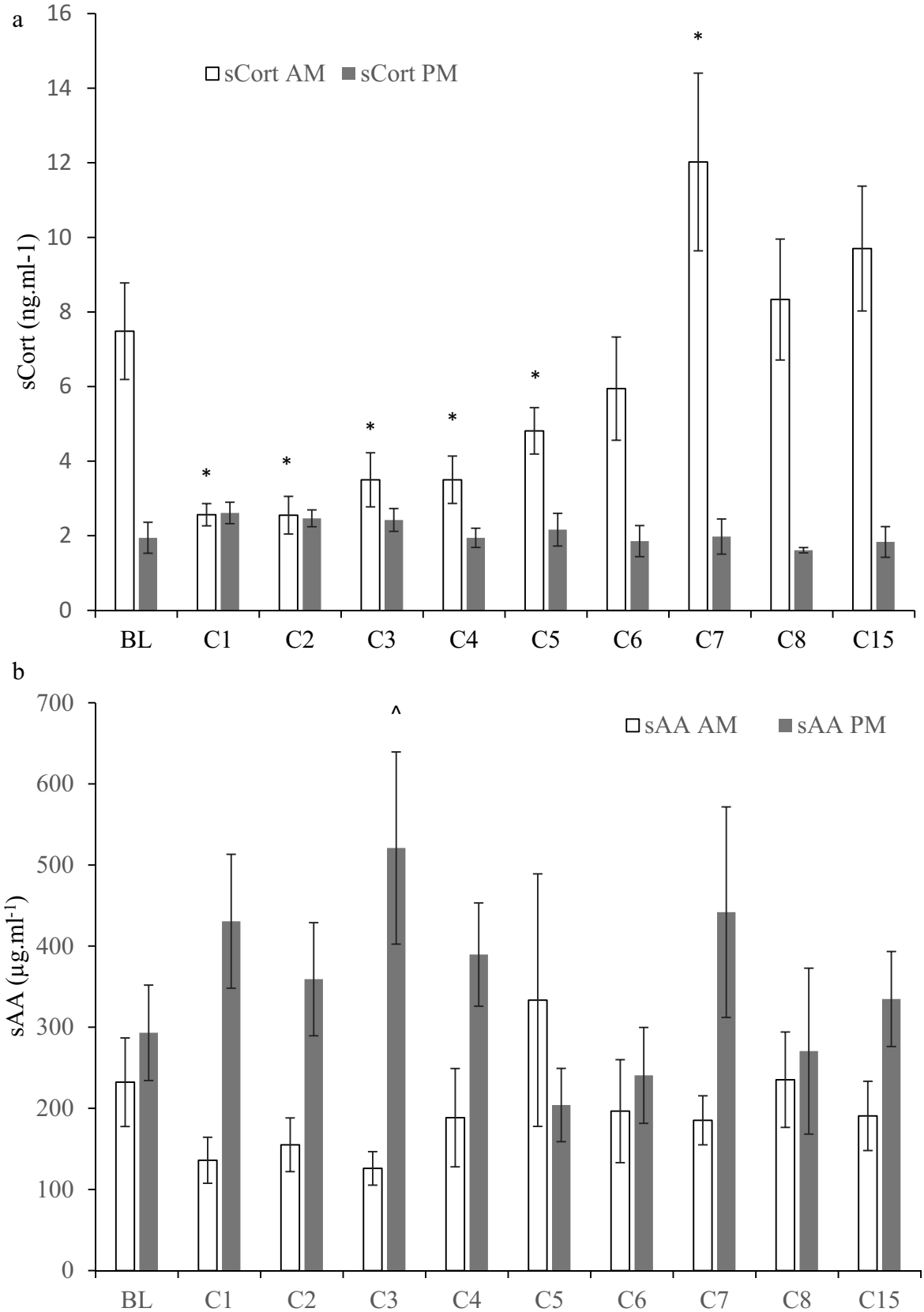

Figure 2. Salivary data. Morning and evening salivary cortical (sCort) (a) and salivary alpha-amylase (sAA) (b) values measured at baseline (BL) and during days 1 to 8 (C1-C8) and day 15 (C15) of pre-Olympic Games camp. Data presented as mean ± SE. * indicates statistical significance ($p < .05$) in comparison to morning scores at BL determined by multi-level regression analysis. ^ indicates statistical significance ($p < .05$) in comparison to evening scores at BL determined by multi-level regression.

Table 4. BRUMS T-scores.

	BRUMS T-score											
	Anger		Confusion		Depression		Fatigue		Tension		Vigour	
	AM	PM	AM	PM	AM	PM	AM	PM	AM	PM	AM	PM
BL	43.9 (0.7)	44.1 (0.9)	44.1 (1.4)	43.5 (0.9)	45.4 (1.6)	45.2 (1.5)	45.1 (2.0)	48.7 (3.3)	48.3 (3.2)	47.7 (2.3)	40.1 (2.6)	37.9 (2.2)
C1	43.0 (0.0)	43.0 (0.0)	44.8 (1.1)	46.9 (2.8)	43.7 (0.4)*	45.1 (1.6)	47.3 (2.6)	53.0 (4.5)	46.1 (2.1)	43.9 (1.4)^	40.9 (3.7)	35.3 (1.6)
C2	43.3 (0.3)	43.5 (0.4)	43.1 (0.6)	51.0 (5.0)^	43.2 (0.1)*	45.8 (1.7)	43.9 (2.6)	60.8 (5.1)^	45.8 (2.0)	46.2 (2.3)	42.8 (3.4)	33.7 (0.9)^
C3	43.0 (0.0)	44.8 (1.1)	44.0 (0.9)	47.9 (3.6)^	45.7 (1.6)	48.0 (2.9)^	49.4 (4.1)	54.9 (3.5)^	43.9 (1.5)*	48.9 (4.0)	42.4 (3.3)	36.8 (1.6)
C4	43.7 (0.4)	45.2 (1.9)	42.2 (0.4)*	44.4 (1.5)	45.1 (1.4)	45.1 (1.4)	47.1 (3.5)	49.0 (3.5)	47.0 (2.9)	43.9 (1.4)^	41.6 (2.9)	37.6 (1.6)
C5	45.4 (2.1)	43.9 (0.8)	43.0 (0.9)	43.6 (1.5)	45.1 (1.7)	44.9 (1.5)	44.6 (3.4)	44.5 (2.3)	44.6 (2.9)*	46.0 (3.0)	42.9 (3.0)	41.0 (2.4)
C6	44.2 (1.0)	43.3 (0.3)	42.7 (0.5)	43.3 (1.0)	45.5 (1.7)	44.3 (1.1)	44.3 (2.3)	51.0 (2.7)	42.8 (1.2)*	44.9 (1.6)	35.8 (1.7)	34.8 (1.2)^
C7	43.0 (0.0)	44.4 (1.3)	41.8 (0.1)*	44.7 (1.7)	44.5 (0.9)	48.9 (2.6)^	41.5 (1.7)	49.4 (3.0)	43.8 (1.5)*	45.0 (1.6)^	47.3 (4.9)*	33.3 (0.8)^
C8	43.4 (0.4)	43.0 (0.0)	41.6 (0.4)*	44.2 (1.8)	45.6 (2.0)	43.8 (0.5)	42.1 (2.3)	42.2 (1.7)^	43.0 (1.5)*	42.7 (1.2)^	39.9 (2.5)	36.2 (1.5)
C15	44.0 (0.9)	43.0 (0.0)	41.6 (0.4)*	43.1 (0.5)	43.9 (0.5)*	43.8 (0.5)	41.4 (1.4)	43.0 (2.0)^	43.0 (0.7)*	42.0 (0.7)^	43.0 (2.0)	38.8 (1.7)

BL, baseline; C, camp; C1 is camp day 1 and so on; * indicates statistical significance ($p < 0.05$) in comparison to BL in the AM; ^ indicates statistical significance ($p < 0.05$) in comparison to BL in the PM; all values are expressed as t-scores; and mean (SE)

Discussion

This study investigated the effects of LHTT on the recovery of Olympic team support staff travelling across eight time-zones in an eastward direction to work at the Pre-Games Camp for the Tokyo 2020 summer Olympic Games. The participants experienced the highest level of perceived jet lag, fatigue, confusion, and lowest sleep duration and efficiency on C2. Feelings of jet lag, higher than normal fatigue, disruption to sleep and bowel function persisted for up to six days following LHTT. In addition, sCort and objective measures of sleep quality and duration were also disrupted for six days post travel. The return of objective and subjective markers to BL, and thus recovery from jet lag and travel fatigue and return of sleep quality and duration, took on average seven days. LHTT also resulted in negative shifts in mood state in comparison to BL, with an increase in fatigue, depression, confusion, and a reduction in vigor reported in the evening times on the days after arrival.

Perceptual markers

Objective measures of circadian disruption are difficult to capture in a free-living environment, thus the present study evaluated jet lag and travel fatigue using self-reporting via the LJMUQ. It was found that on arrival the participants perceived feeling significantly jet lagged and fatigued for six days post-travel, which is most likely attributed to circadian disruption due to crossing time-zones in an easterly direction. The participants experienced elevated fatigue in the evening times only; with significant elevations in comparison to BL reported on C1, C2 and C6. This is surprising, considering it has been reported previously (Janse van Rensburg et al. 2021; Roach and Sargent 2019) that following rapid travel across eight time-zones in an eastward direction that an individual is likely to find it difficult to initiate sleep.

In this group of Olympic team support staff, the most severe perceived jet lag and fatigue was experienced on C2. The human body clock needs time to gradually adjust and consequently the symptoms of jet lag will persist until circadian rhythms shift to a new environment. It took seven days for perceived jet lag and fatigue to normalize; which was aligned with the expected time-frame of "complete" jet lag recovery of one day for every time-zone crossed (Eastman and Burgess 2009; Janse Van Rensburg et al. 2020; Loat et al. 1989; Roach and Sargent 2019). There were occasions of disruption to mental performance, appetite, and bowel function throughout the study period, but it cannot be determined if they are related to jet lag or travel fatigue, or due to factors unrelated to travel, such as work program demands, environmental factors, or cultural differences.

Sleep

Sleep duration and sleep efficiency data collected by actigraphy in the week prior to LHTT indicated that sleep was less than optimal in the present group of participants. At BL, support staff spent on average 8.3 h in bed and got 6.8 h sleep. A minimum sleep duration of 7.0 h is recommended for healthy adults aged from 18 to 64 y by the National Sleep Foundation (Hirshkowitz et al. 2015). At BL, average sleep latency of 33.3 min and sleep efficiency of 83.4% were recorded. Normal adult sleep latency is expected to be between 10 and 20 min (Thomas and Anderson 2013) and a sleep efficiency value of <85% is commonly used as a clinical and research cut-off for identifying insomnia (Markwald et al. 2016). A recent consensus statement on the management of travel fatigue in athletes (Janse van Rensburg et al. 2021) recommended to "protect sleep" and "be well-rested" in the pre-travel period. The present findings indicate that sleep was suboptimal and casts doubt on whether this group of support staff were well-rested

in the days before travel. Thus, it would be beneficial to extend similar recommendations on sleep pre-travel to support staff travelling with athletes, as there may be potential to reduce the effects of travel fatigue and expediate recovery following LHTT.

On the night of arrival, the participants of this study had greater propensity to sleep, with a bedtime of over 2 h earlier and shorter sleep latency in comparison to BL. These findings are supported by previous studies on healthy individuals (Nicholson et al. 1986) and elite athletes (Rossiter et al. 2022), suggesting that on the night of arrival, the sleep debt accumulated during travel was enough to override any wakefulness in the evening time caused by circadian misalignment. However, get up time was also over 2 h earlier than BL, resulting in similar sleep duration and efficiency on C2.

Data recorded by actigraphy indicated that in comparison to BL, these participants experienced disruption to sleep schedule following LHTT, particularly between C2 and C6. Participants went to bed significantly earlier on some nights, however also got up earlier every morning, at times up to 2.5 h earlier than BL. Subsequently, time in bed, sleep duration and perceived sleep was significantly worse than BL on most mornings between C2 and C6. In comparison to BL, sleep duration and efficiency normalized on C7. This is slightly longer than the expectation the of "partial" jet lag recovery in the form of improved sleep duration and quality occurring 4–5 days after eastward LH travel across eight time-zones (Roach and Sargent 2019), when the circadian system shifts sufficiently so that minimum core body temperature occurs during the main night-time sleep period. In a study on flight attendants (Haurma et al. 1994) it was found that following 4-day round trip flights over 10 time zones that it took 4 days for sleep duration and quality to return to pre-flight values. In a study on healthy males (Nicholson et al. 1986) it was found that sleep duration and quality was still disturbed on the fifth night after an eastward flight across five time-zones. A recent study (Rossiter et al. 2022) examining the effects of eastward travel across eight time-zones on sleep in international swimmers found that on average it took five days to normalize. The reason for slower than expected normalization of sleep duration is unclear. It should be considered that the social and work activities of the staff were not controlled or monitored during the study period thus it is plausible that the sleep schedules may have at least been partially influenced by their work/social schedules and demands. These findings are supported by a case study comparing sleep in a head coach to his team, which similarly observed that the sleep/wake behavior of the coach was poorer than the athletes following LHTT for major competition, particularly on the nights before important games (Lastella et al. 2017).

Sleep actigraphy data showed that, except for one night, sleep latency tended to be shorter and bedtime earlier on the camp in comparison to at home, indicating that the participants did not have difficulty initiating sleep at their normal bedtime. This is supported by LJMUQ and mood profiling finding higher fatigue and lower vigor levels in the evening time. This is contrary to the expectation that following an eastward flight across eight time-zones that individuals will have difficulty initiating sleep at their regular bedtime, and they will sleep-in later than usual (Janse van Rensburg et al. 2021; Roach and Sargent 2019) but supports findings of (Rossiter et al. 2022) who found that following similar LHTT, international swimmers awoke significantly earlier on the days after arrival. Again, it is plausible that the sleep/wake cycle may have been influenced by exogenous factors independent of the body clock (Waterhouse et al. 2007). It is possible that work commitments in the morning times, at the nadir of the circadian cycle following eastward flight across eight time-zones, led to an accumulation of fatigue in these participants which superseded the peak of the circadian (Rossiter et al. 2022) cycle in the evening time, resulting in earlier bedtimes and shorter sleep latency in comparison to BL. Early awaking observed in this study and others may indicate, that the circadian sleep-wake cycle may become a greater influence in the early morning hours leading to inappropriately timed wakefulness during the second half of the night as is thought to occur in insomnia (Espie 2002; Vosko et al. 2010).

By the end of this study (C15) perceived sleep normalized. Although get up times remained significantly earlier than BL, total sleep time was significantly extended to 7.6 h, sleep latency was reduced to 16.5 min and sleep efficiency was 87%, all within expected ranges for healthy adults (Hirshkowitz et al. 2015; Markwald et al. 2016; Thomas and Anderson 2013).

Physiological markers

Cortisol is considered a robust objective chronobiological marker (Spencer et al. 2018), following a predictive diurnal rhythm with relatively high values in the morning time, followed by a decline throughout the day reaching the lowest point at around midnight (Pruessner et al. 1997). This study used the LFD method for analyzing cortisol in saliva which has been deemed valid (0.52) and reliable (r = 0.69) for measurement of sCort (Fisher and Sinclair 2015). In the present study, sCort AM and PM concentrations at BL were in good

agreement with those previously published (Aardal and Holm 1995). This study found that on the morning after travel (C1), there was a 66% decrease in sCort AM in comparison to BL, so that sCort AM on C1 was similar to sCort PM at BL. Similar changes in cortisol concentrations have been observed following eastward travel across seven time-zones in healthy individuals (Piérard et al. 2001) and eight time-zones in elite athletes (Rossiter et al. 2022); thus confirming that LHTT induces desynchronization of cortisol secretion. sCort AM remained depressed in comparison to BL until C5, and normalized on C6, coinciding with the alleviation of perceived jet lag from C7 onwards. This suggests that sCort is a convenient and cost-effective objective method to indicate resynchronization of circadian rhythm following LHTT.

Unexpectedly on C7, sCort AM increased to significantly exceed BL. Although it is possible that this response is due to psychological or physiological stress, it is unlikely, because sCort is much more responsive to acute stress in the evening time in comparison to the morning (Spencer et al. 2018). sCort was similar in the evening time in comparison to BL throughout the study period, including C7. Elevated sCort on the morning of C7 coincides with extended sleep duration in comparison to the six nights previous and significantly elevated "vigor" as reported by the BRUMS. Short-term sleep deprivation (3 h sleep/night for 4 consecutive days) has been shown to inhibit the activation of the HPA-axis and reduce serum cortisol levels and an increase in perceived stress (Wu et al. 2008) and positive associations have been found between longer sleep duration and single measures of sCort at awakening ($r = .469$, $p < .001$, Stalder et al. 2009). Therefore, it is conceivable that sleep restoration that occurred on the night of C7 caused an elevation of sCort and resulted in improved vigor on the following morning.

Production of alpha-amylase (AA) is regulated by the autonomic nervous system and is strongly influenced by physical and psychological stress (Chatterton et al. 1996). In elite soccer players, IPRO has shown good agreement with the ELISA method for measurement of sAA ($r = 0.87$; 95%CI = 0.78–0.92) and presented a coefficient of variation for replicate measures of 13.2% (Dunbar 2015).

Consistent with previous research (Nater and Rohleder 2009), this study found that BL sAA concentration has a diurnal rhythm with lowest sAA values observed in the morning and highest values in the evening. This rhythm persisted on the days following LHTT except for C5, where an elevation of sAA resulted in the AM value to be greater than the PM value, however this was insignificant ($p = .07$). On the days following LHTT, sAA concentrations were similar to BL in both the AM and PM, apart from the evening of C3. Elevated sAA concentrations on C3 (PM) and C5 (PM) may indicate an acute elevation in psychological stress experienced by an individual in response to an incident, rather than related circadian disruption of physical and psychological stresses associated with LHTT.

Mood

Moods are influenced by situational factors such as interactions with other individuals and environmental forces (Terry Peter and Lane 2011). The present study used the BRUMS (Terry and Lane 2010) to assess mood states in the participants before and after LHTT. The BRUMS has been validated for diverse adult populations and cultures (Canadas et al. 2017; Han et al. 2020; Zhang et al. 2014) and contexts (Fazackerley et al. 2003; van Wijk et al. 2013) including LHTT (Terry 2010). The BRUMS has demonstrated satisfactory internal consistency, with Cronbach alpha coefficients ranging from 0.74 to 0.90 for the six subscales (Terry et al. 1999).

This study found that following LHTT in an easterly direction, there was evidence of a negative shift in mood state in the evening time in comparison to BL. Conversely, mood profiles tended to be more positive in the morning times. This provides further evidence that contrary to the expectation that the evening time would present the peak of the circadian cycle following eastward flight across eight time-zones, the evening times were unlikely yield optimum performance in this group of support staff. This suggests that following LHTT in an eastward direction, it would be beneficial to schedule key support and decision-making tasks in the morning time when individuals are more likely to experience reduced fatigue and positive mood states.

Similar findings of reduced vigor and increased fatigue, identified as a "shark fin" mood profile (Terry and Parsons-Smith 2021), following LHTT have been previously associated with jet lag and travel fatigue experienced by elite athletes (Hill et al. 1993; Terry 2010; Terry Peter and Lane 2011). In addition, the present study found elevated confusion and depression scores in these Olympic team support staff. Increased confusion has also been observed in a previous study involving healthy adults following LHTT (Hill et al. 1993), suggesting that BRUMS might be a useful tool to use to assess mood disturbances caused by jet lag and travel fatigue.

Despite elevated jet lag and fatigue on arrival, mood profiling indicated that arrival to the pre-Games holding camp was associated with some positive changes to mood state particularly with regards reduced tension in comparison to BL. Previously (Hill et al. 1993)

described an "iceberg" mood profile in healthy individuals and elite athletes in the days leading up to LHTT. However, it has been found (Pedlar et al. 2007) found that tension in a female explorer travelling to the south pole tended to increase as the expedition drew nearer. In the time leading up to an Olympic Games there can be an intensification of stressors, such as the logistics of transporting a large team, demands from family, friends and media, transport of essential equipment, cultural differences, Olympic Games accreditation etc. (Terry et al. 1997). It also presents a situation of extended time away from family, friends, and other personal and professional commitments. Additionally, in the present study there was the unprecedented situation of an Olympic Games being delayed by one year due to Covid-19 and being held amid a global pandemic, which is likely to have presented additional concern to Olympic team support staff. It is also possible that sub-optimal sleep quality and duration experienced by these individuals in the days leading up to LHTT effected mood (Dinges et al. 1997). Positive changes to mood profiles occurred in the morning times following LHTT, despite further reductions in sleep duration and quality, thus is likely to be reflective of alleviation of stress and uncertainty on the safe arrival to their destination.

Mood profiling indicates that on C8 and C15 the participants in this study experienced improved fatigue score in comparison to BL. However, vigor levels remained low, possibly indicating a "submerged" mood profile (Parsons-Smith et al. 2017). This was contrary to observations by Terry's (2010) mood profiling of an Olympic athlete on the days following LHTT; in this case study, it was found that despite low vigor and high fatigue levels being apparent on the days following LHTT; by day 17 post-travel the athlete's mood profile had transitioned to an "iceberg" profile typically associated with superior performance. A case study (Terry and Parsons-Smith 2021, 8) describes an athlete with a submerged profile as "feeling flat and emotionally drained from inter-continental travel and a period of hard physical training but represented no threat to mental wellbeing." Little is known about the behavioral and mental health of a submerged profile (Terry and Parsons-Smith 2021); however, it is thought to be characterized by being relatively devoid of emotion, including vigor, or feeling "flat" and possibly impede goal-directed behavior in a variety of contexts (Parsons-Smith et al. 2017), which may be of concern to Olympic team staff who are supporting athletes in their final preparation for the Olympic Games.

These findings suggest that BRUMS might be a useful tool not only to track recovery from jet lag and travel fatigue in support staff working with elite athletes; but it may also be used to support emotional wellbeing of support staff and the appropriate implementation of mood management strategies before and after LHTT and particularly in the approach to major competition, such as an Olympic Games.

Study limitations

Given the ecological nature of data collection, certain limitations should be acknowledged. The number of participants in this study was limited because of the competition level and prohibitive cost of transporting individuals on an LH flight. Due to restrictions in place to prevent the spread of Covid-19, researchers did not directly supervise the completion of questionnaires and collection of salivary samples, which may have affected compliance and standardization of procedures for some variables. To ensure standardization of procedures, a one-to-one online video session and detailed written instructions on data collection procedures were provided to each participant before the study commenced. To enhance compliance, during the study period reminder text messages were sent to each participant on the evening before each data collection point.

The lack of control for work/social schedules and activities may have influenced some of the variables in this study. Furthermore, BL measurements were taken over a two-day period in the week before travel and evidence suggests that this group of individuals may have been experiencing high levels of tension and sub-optimal sleep during this time. However, since these factors are not usually controlled for in real-life, our results can be considered ecologically valid.

As noted previously (Waterhouse et al. 2002), this study found a high degree of individual variability in the types and severity of symptoms associated with jet lag and travel fatigue and varying rates of recovery following LHTT. Characteristics that contribute to this variation are sex, age, chronotype, and travel experience (Bin et al. 2019). Furthermore, circadian rhythm is regulated by the master circadian clock located in suprachiasmatic nuclei (SCN) of the hypothalamus, however there are clocks in every cell of the body, as well as peripheral clocks located in all tissues and organs. Following LHTT, there is disorganization between the clocks in different organs and tissues, and it is expected that resynchronization occur at different rates (Revell and Eastman 2012). It has been shown that following a 6-hour phase advance in mice that the SCN resynchronized by day 3, in comparison to >8 days for peripheral tissues (Davidson et al. 2009). Other possible reasons are variations in travel schedules, exposure to light, timing of physical activity, sleep, and diet; all of which have been described as possible interventions for jet lag and

travel fatigue (Janse Van Rensburg et al. 2020; Roach and Sargent 2019; Waterhouse et al. 2002). The participants in the present study received travel, sleep and light exposure guidelines designed specifically for their journey and followed similar meal and sleep schedules, however no attempt was made by the researchers to control these variables.

Practical implications

This study presented a unique opportunity to observe the effects of LHTT for major competition on Olympic team support staff. Our results present a detailed insight on perceived jet lag and travel fatigue, mood profiles, sleep duration and quality, physiological markers for circadian rhythm and stress in study participants during this time. This study highlights that LHTT is a source of physical and psychological stress to Olympic team support staff and provides a foundation for future research to assist support staff and agencies for elite sport to better understand the stressor of LHTT for major competition and its effects on circadian rhythms, sleep, and mood. It identifies that support agencies should strive to provide education and financial investment to facilitate their staff to implement strategies and coping mechanisms to reduce the effects of travel fatigue and jet lag, to expediate recovery from LHTT on arrival to their destination and to manage the psychological stress associated with LHTT for major competition. Recommended strategies specific to support staff embarking on LHTT for major competition are the implementation of mood regulation strategies and sleep hygiene before and after LHTT. Management of work programs is required before LHTT to allow support staff to commence their journey in a well-rested state and following LHTT to allow for adequate recovery. It is recommended that, where possible, support staff arrive to the destination in advance of the athletes/team and key support and decision-making tasks are scheduled at times of day where support staff perceive they can perform optimally. These findings and recommendations should help facilitate future Olympic support teams to accelerate recovery from LHTT and to enhance the quality of their Olympic Games experience. Ultimately this will result in optimizing their performance and support to athletes in final preparations for the Olympic Games and to enhance athlete performance at the Olympic Games.

Conclusion

In conclusion, this novel study is the first to provide a comprehensive presentation and evaluation of ecologically valid perceptual, mood, sleep, and physiological data from Olympic team support staff before and after LHTT.

It took seven days for support staff to perceive themselves to be fully recovered from jet lag and for sleep duration and quality to return to BL following LHTT in an eastward direction across eight time-zones. This was preceded by normalization of sCort, a physiological marker of circadian rhythm, six days post-travel. This study found that LHTT did not result in change in physiological markers associated with psychological stress. Although there were negative shifts in mood states following LHTT, with elevated fatigue, confusion and depression and reduced vigor in the evening times, arrival to Japan also resulted in positive changes in mood states in these participants with some alleviation of tension. Towards the end of the study period, although jet lag and travel fatigue had dissipated, and sleep quality and duration improved, the participants displayed "submerged" mood state where vigor remained low.

The data and findings from this study indicate that Olympic team support staff engaging in LHTT in an eastward direction across eight time-zones should pay attention to their work program during the first seven days after arrival to allow adequate recovery from jet lag and travel fatigue. There is suggestion that special consideration should be given to strategies to improve sleep and assist with mood regulation before and after LHTT for the Olympic Games.

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